

Types of Differential Eqn

① Ordinary Differential Eqn, ODE

Egns where the original derivatives are taken w respect to only one variable. That is, there is only one independent variable.

$$f(t) = m \frac{d^2x}{dt^2} + c \frac{dx}{dt} + kx$$

② Partial differential eqns, PDE

Egns that depend on partial derivatives of several variables. That is, there are several independent variables.

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$$

③ Linear & non Linear ODE [Properties b/w are for linear]

- ① The dependent variable y and its derivatives power are of 1st degree
- ② Each coefficient depends only on the independent variable x
- ③ y cannot be any non-linear function [e^y , $\sin(y)$...]

Types of problems

① Initial value problem, IVP

ODE with initial conditions

$$\frac{d^2z}{dt^2} = -g \quad ; \text{ Given } z = 1, \frac{dz}{dt} = 10, \text{ at } t = 0 \text{ s}$$

② Boundary value problem, BVP

ODE with boundary conditions

$$\frac{d^2z}{dt^2} = -g \quad ; \text{ Given } z = 0 \text{ at } t = 0 \\ z = 3 \text{ at } t = 0$$

Type of solutions

① General Solution

Solution involving an arbitrary constant

② Particular Solution

Solution with arbitrary constant determined with initial conditions / boundary conditions

1st order separable ODEs

Separable

$$\frac{dy}{dx} = [f(x)][g(y)]$$

Example: $\frac{dy}{dx} = e^{x+y}$

$$\frac{dy}{dx} = (e^x)(e^y)$$

Not separable

(can't be written in $\frac{dy}{dx} = f(x)g(y)$ form.)

Example: $\frac{dy}{dx} = \frac{x+y}{x}$
 $= 1 + \frac{y}{x}$

Finding a solution for separable ODE

$$\frac{dy}{dx} = [f(x)][g(y)]$$

$$\frac{1}{g(y)} \frac{dy}{dx} = [f(x)]$$

$$\int \frac{1}{g(y)} \frac{dy}{dx} dx = \int f(x) dx$$

$$\boxed{\int \frac{1}{g(y)} dy = \int f(x) dx}$$

Example 1:

$$\frac{dy}{dx} + \frac{x}{y} = 0$$

$$\frac{dy}{dx} = -\frac{x}{y}$$

$$\int y \frac{dy}{dx} dx = -\int x dx$$

$$\int y dy = -\frac{x^2}{2}$$

$$\frac{1}{2}y^2 = -\frac{x^2}{2} + C$$

Example 2:

$$\frac{dv}{dt} = -k\sqrt{v}, \quad v = v_0 \text{ at } t = 0$$

$$\int v^{-1/2} dv = -\int k dt$$

$$2v^{1/2} = kt + C \quad [\text{General Solution}]$$

When $v = v_0$, $t = 0$

$$2v_0^{1/2} = C$$

$$\therefore 2v^{1/2} = kt + 2v_0^{1/2}$$

$$v = \left[v_0^{1/2} - \frac{1}{2}kt \right]^2 \quad [\text{Particular solution}]$$

Finding solution for 1st order linear ODE

$$\frac{dy}{dx} + p(x)y = Q(x)$$

Notice that $\frac{d}{dx} [y e^{\int p(x) dx}] = e^{\int p(x) dx} \frac{dy}{dx} + e^{\int p(x) dx} p(x) y$

Integrating Factor, $I(x)$

$\therefore$ When we multiply $e^{\int p(x) dx}$ to the 1st order ODE, we get

$$e^{\int p(x) dx} \frac{dy}{dx} + e^{\int p(x) dx} p(x) y = [Q(x)] e^{\int p(x) dx}$$

$$\frac{d}{dx} [y e^{\int p(x) dx}] = Q(x) e^{\int p(x) dx}$$

Applying $\int dx$ to both sides,

$$y e^{\int p(x) dx} = \int [Q(x)] [e^{\int p(x) dx}] dx$$

$$y = \frac{1}{I(x)} \int [Q(x)] [I(x)] dx$$

Example 1:

$$\cos(x) \frac{dy}{dx} - y \sin x = x^2$$

$$\frac{dy}{dx} + \left(\frac{-\sin x}{\cos x} \right) y = \frac{x^2}{\cos(x)} \quad \text{--- (i)}$$

$$\text{Let } I = e^{\int \frac{-\sin x}{\cos x} dx} = e^{\ln \cos x} = \cos x$$

Multiplying both sides of (i) with I ,

$$\implies (\cos x) \left(\frac{dy}{dx} \right) + (-\sin x)(y) = x^2$$

$$\frac{d}{dx} [\cos(x) y] = x^2$$

$$\cos(x) y = \int x^2 dx$$

$$y = \frac{1}{\cos x} \left[\frac{x^3}{3} + C \right] \neq$$

Solving 1st order, linear, constant-coefficient, homogeneous ODE

$$\frac{dy}{dx} + \lambda y = 0$$

$$\left(\frac{1}{y}\right) \frac{dy}{dx} = -\lambda$$

$$\int \frac{1}{y} dy = -\lambda \int 1 dx$$

$$\ln |y| = -\lambda x + c$$

$$y = \pm e^{-\lambda x + c}$$

$$= (\pm e^c)(e^{-\lambda x})$$

$$= A e^{-\lambda x} \quad \text{where } A = \pm e^c$$

Example 1:

$$5 \frac{dy}{dx} + y = 0 \quad \text{--- (1)}$$

Since this is a 1st order, linear, constant-coefficient, homogeneous ODE, we know for sure $y = A e^{\lambda x} \implies \frac{dy}{dx} = \lambda A e^{\lambda x}$

$$\therefore \text{(1) becomes } 5[A\lambda e^{\lambda x}] + A e^{\lambda x} = 0$$

$$A e^{\lambda x} [5\lambda + 1] = 0$$

$$5\lambda + 1 = 0 \quad [\text{Auxiliary Eqn}]$$

$$\lambda = -1/5$$

$$\therefore \text{General solution : } y = A e^{-\frac{1}{5}x} *$$

* Note

$$\text{Let } \frac{dy}{dx} + k y = 0, \text{ and we know } y = A e^{\lambda x} \implies \frac{dy}{dx} = \lambda A e^{\lambda x}$$

$$\therefore \lambda A e^{\lambda x} + k A e^{\lambda x} = 0$$

$$\lambda + k = 0 \quad [\text{Auxiliary Eqn}]$$

$\therefore$ We can immediately write the auxiliary Eqn by changing
 $\frac{dy}{dx} \rightarrow \lambda$, $y \rightarrow 1$

Solving 1st order, linear, constant-coefficient, ODE

$$\frac{dy}{dx} + \lambda y = Q(x) \quad \text{--- ①}$$

Let $I = e^{\int \lambda dx} = e^{\lambda x}$,

Multiplying both sides of ① by I

$$\Rightarrow e^{\lambda x} \frac{dy}{dx} + e^{\lambda x} \lambda y = [Q(x)] [e^{\lambda x}]$$

$$\frac{d}{dx} [e^{\lambda x} y] = [Q(x)] [e^{\lambda x}]$$

$$y = \frac{1}{e^{\lambda x}} \int [Q(x)] [e^{\lambda x}] dx$$

$$= \boxed{} + c e^{-\lambda x}$$

This term is the $\int [Q(x)] [e^{\lambda x}] dx$ term without the arbitrary const.
[Particular Integral]

This appears no matter what is $Q(x)$.
[This always satisfies the eqn when $Q(x) = 0$, as shown in the previous pg.]
[Complementary function]

$$\therefore y = y_p + y_c$$

(
Particular
Integral)

Complementary
Function, $A e^{\lambda x}$

Right-hand side (i.e. the $f(x)$)	Trial function
Linear function	$Px + Q$
Polynomial of degree n	$P_n x^n + P_{n-1} x^{n-1} + \dots + P_1 x + P_0$ (typically we will just use letters P, Q, R , etc rather than using subscripts on the same letter)
Trigonometric function involving $\cos kx$ and/or $\sin kx$	$P \cos kx + Q \sin kx$
Exponential function involving e^{kx}	$P e^{kx}$
Sum of different functions	Sum of matching functions

Example 1:

$$\frac{dy}{dx} + 2y = x^2$$

$$\text{Auxiliary Egn : } \lambda + 2 = 0$$

$$\lambda = -2$$

$$\text{Complementary func., } y_c = Ae^{-2x}$$

$$\text{Particular integral, } y_p = ax^2 + bx + c$$

$$y_p' = 2ax + b$$

$$\therefore (2ax + b) + 2(ax^2 + bx + c) = x^2$$

$$(2a)x^2 + (2a + 2b)x + (b + 2c) = x^2$$

$$\therefore 2a = 1 \Rightarrow a = \frac{1}{2}$$

$$2\left(\frac{1}{2}\right) + 2b = 0 \Rightarrow b = -\frac{1}{2}$$

$$-\frac{1}{2} + 2c = 0 \Rightarrow c = \frac{1}{4}$$

$$\therefore y_p = \frac{1}{2}x^2 - \frac{1}{2}x + \frac{1}{4}$$

$$\begin{aligned} \therefore y &= y_c + y_p \\ &= Ae^{-2x} + \frac{1}{2}x^2 - \frac{1}{2}x + \frac{1}{4} \quad \text{H} \end{aligned}$$

* Note that we could also use the Integrating factor method to solve the eqn.

1st order ODE

Separable

$$\frac{dy}{dx} = f(x) g(y)$$

Rearrange & Integrate

$$\frac{1}{g(y)} dy = f(x) dx$$

Linear

$$\frac{dy}{dx} + P(x) y = Q(x)$$

Integrating factor

$$I = e^{\int P(x) dx}$$

$$\frac{dy}{dx} = F\left(\frac{x}{y}\right)$$

$$\text{Let } u = \frac{y}{x}$$

$$\text{Sub } y = ux, \frac{dy}{dx} = v + x \frac{dv}{dx}$$